CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 1

Answers

CANDIDATE NAME	
CLASS	2T
INDEX NUMBER	

BIOLOGY

8876/02

Paper 2 STRUCTURED & FREE RESPONSE QUESTIONS

20 AUGUST 2018
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1 [8]	
2 [12]	
3 [8]	
4 [5]	
5 [12]	
6 or 7 [15]	
TOTAL P2	60

This document consists of **25** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** questions in this section

1 **(a)** Fig. 1.1 is an electron micrograph showing part of an organelle present in a mature plant cell.

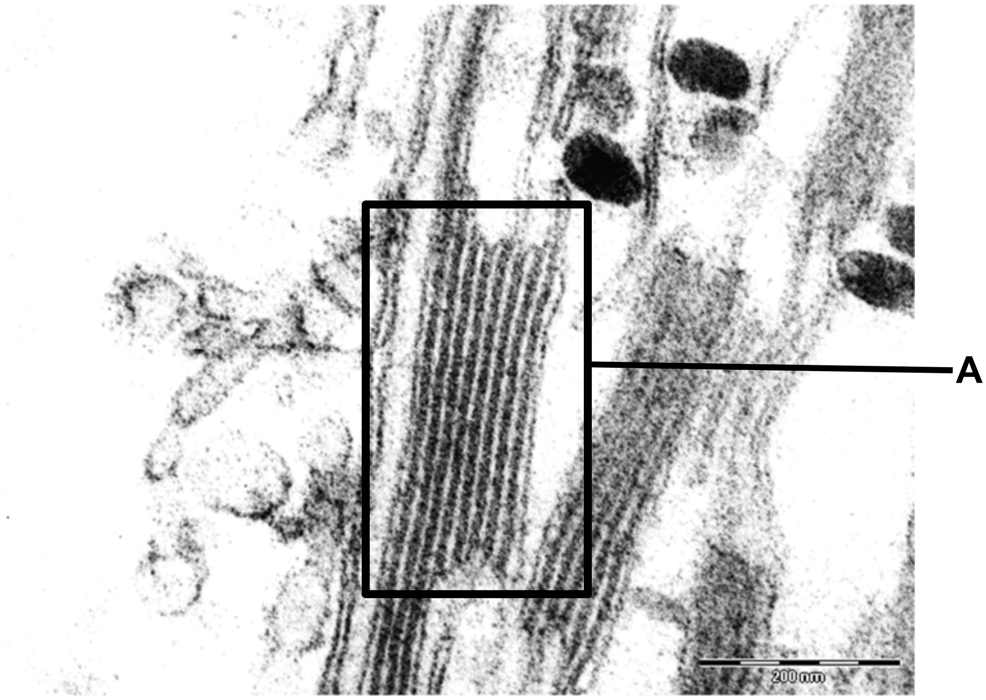


Fig. 1.1

(i) Identify the organelle in which structure **A** resides.

....[1]

ANS [L1] Novel [1]

SC: Identify	**organelle**	**cell structure A**
OR:	**Chloroplast**	

1. Chloroplast

(ii) Explain the significance of the flattened stack arrangement in structure **A**.

....

....

....

....[2]

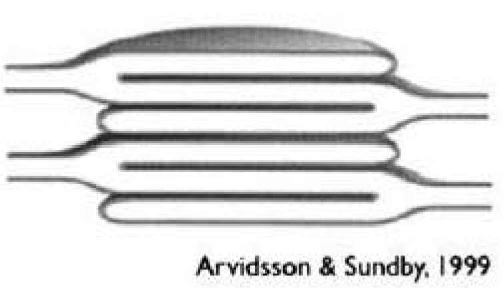
ANS [L2] Novel [2]

SC: explain	**significance**	**flattened stacks A**
OR:Give reasons	**importance**	**1] increase SA; maximise attachment of photosynthetic pigments max light absorption**
		2] increase thylakoid space ❼ build up proton grad

The thylakoids are arranged in flattened sacs in stacks

1. is to provide **large surface area** to maximize the **attachment of** (photosynthetic elements) / electron protein carriers, ATP synthases and **photosynthetic pigments** for **maximum light absorption**;

2. is to **increase** thylakoid **space** for the **building up of proton gradient** needed for ATP synthesis via chemiosmosis;



(b) Fig. 1.2 shows the structure of a cell surface membrane.

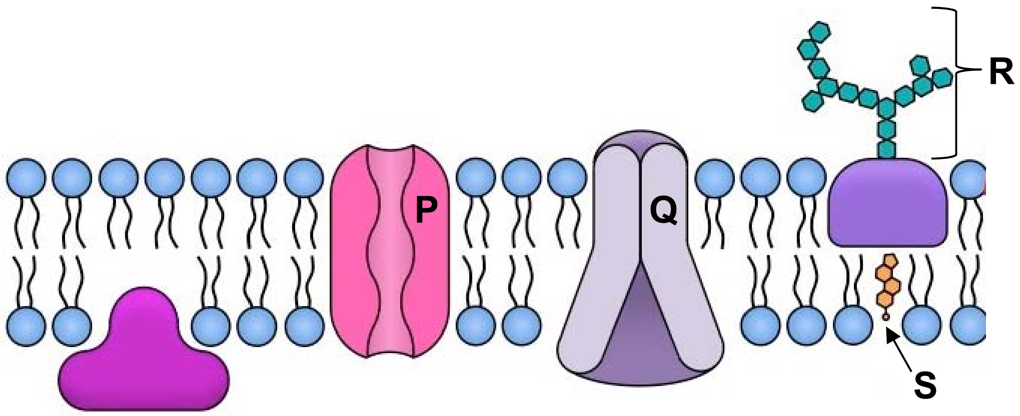


Fig. 1.2

With reference to Fig. 1.2,

(i) explain how the structure of **Q** allows it to be embedded in the cell surface membrane.

....

....[1]

ANS [L1] Novel [1]

SC: Explain how	**structure Q**	**embedded**
OR:		**made up of both hydrophilic and hydrophobic R-groups A.A**

1. Structure Q / Carrier protein Q / Q has 3D conformation shape that are made up of **both hydrophilic and hydrophobic amino acids** that folds and anchored on the cell membrane where it can form hydrophilic and hydrophobic interactions with the phospholipid bilayer respectively / OWTTE

(ii) explain how structure **P** functions differently compared to structure **Q**.

....

....[1]

ANS [L2] Novel [1]

SC: explain how	**P**	**Q**	**function differently**
OR:	**(channel)**	**(carrier)**	

1. P allows substances to pass through cell membrane via **facilitated diffusion** only whereas Q allows both **facilitated diffusion and active transport**.

(iii) explain why structure **S** is amphipathic.

....

....[1]

ANS [L2] Novel [1]

SC: Explain why **structure** **S** **amphipathic**
OR: **S: cholesterol: 4 fused rings + hydrocarbon tails ❼ hydrophobic; -OH group ❼ hydrophilic**

1. **S / cholesterol** contains both a **hydrophilic –OH group** and a **hydrophobic 4 fused C rings** that wedged neatly between the phospholipid bilayer.

An immune response is triggered when the body recognises and defends itself against any substances that appear foreign to the body.

(iv) Suggest why a change in R may cause an immunological response.

....

....[2]

ANS [L3] Novel [2]

SC: suggest why **a change in R** **immunological response**
OR: **R: R ❼ function for cell recognition**
Change/Mutated glycoprotein R ❼ changes on CBH chains ❼ cannot recognise as self / see as foreign ❼ trigger immune response

1. **Glycoprotein R** plays an important role for **cell-to-cell recognition** / **serve as identification tags**
2. With **mutated/changes** in **glycoprotein R**, this **changes** the types of sugar residues that make up the **carbohydrate chains** on glycoprotein R which is **considered foreign / cannot recognise as self** thus triggering an immune response.

[Total: 8 marks]

2 **(a)** Fig.2.1 shows the structure of an adult haemoglobin molecule, HbA.

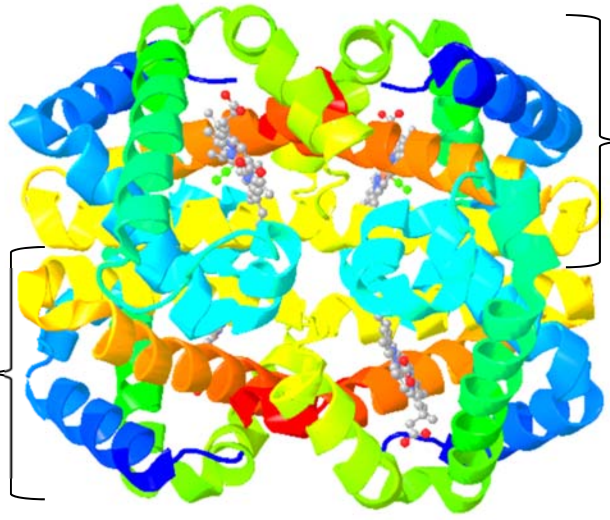


Fig. 2.1

(i) With reference to Fig. 2.1, explain how the structure of HbA is related to its function.

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....[2]

ANS [L1] (Novel) [2]

SC: Explain how **structure of HbA** **function**

OR:

	STRUCTURE	**FUNCTION**
1	HbA molecule has an **overall globular shape** that allows **compact packing**	Allows **many HbA molecules to be packed into a red blood cell** for transport of oxygen.
2	**4 subunits in each haemoglobin molecule**. **Each subunit in haemoglobin consisting of a protein (globin) and a prosthetic (non-protein) haem group component.**	This allows haemoglobin to bind to four oxygen molecules, which greatly **facilitates the transport of oxygen** by haemoglobin.
4	Four subunits in each haemoglobin molecule, **each consisting of a protein (globin) and a non-protein (haem group) component.**	This is the haem binding site. It **is lined with hydrophobic amino acid residues** to provide a hydrophobic environment for the haem group, which is largely hydrophobic.
5	The haem group consists of a porphyrin ring and an iron ion (Fe^{2+}).	**Allowing the release of oxygen in metabolically active tissues such as muscle.**
6	Within haemoglobin, **each polypeptide (α subunit and β subunit)** consists of **both hydrophilic and hydrophobic amino acid residues**. The bulk of **the hydrophobic amino acid residues are buried in the interior of the globular structure** while the **hydrophilic amino acid residues are on the outside**.	This makes the haemoglobin **soluble** in aqueous medium allowing haemoglobin to be a **good transport protein for oxygen** in blood.

Max 2

Sickle cell anaemia is an inherited disease due to a mutation in the allele that codes for β-globin polypeptide chain. This mutant HbS allele results in the production of haemoglobin variant HbS.

Figure 2.2 shows the blast sequence of both the normal adult haemoglobin variant, HbA, as well as the mutant adult haemoglobin variant, HbS.

Beta globin sequence in normal adult haemoglobin (HbA)

nucleotide base	C T G A C T C C T G A G G A G A A G T C T
amino acid	Leu – Thr – Pro – Glu – Glu – Lys – Ser

Beta globin sequence in mutant adult haemoglobin (HbS)

nucleotide base	C T G A C T C C T G T G G A G A A G T C T
amino acid	Leu – Thr – Pro – Val – Glu – Lys – Ser

Fig. 2.2

(ii) With reference to Fig.2.1 and Fig. 2.2, explain how this mutation affects the structure of the haemoglobin molecule.

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....[2]

ANS [L2] (Novel) [2]

SC: Explain how **affect structure of haemoglobin mol.**

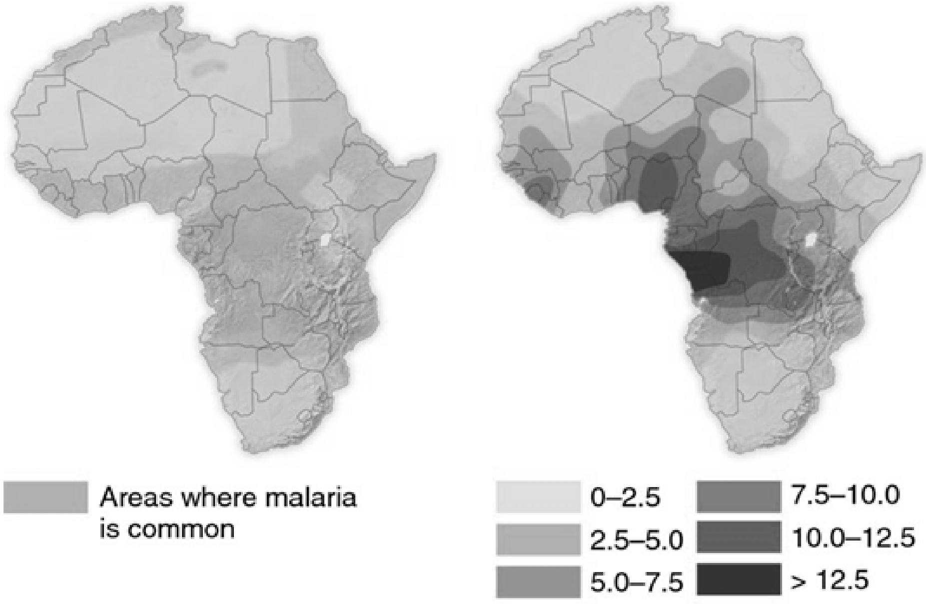
OR:
1. single-base substitution; T ❼ A and mRNA codon from GAG ❼ GUG
2. Glu (hydrophilic/polar) ❼ Val (hydrophobic/nonpolar) ❼ change 3D conf. ❼sickle shaped rbc

1. Due to **single-base substitution, thymine** in **normal** DNA sequence is **replaced** with **Adenine** in the **mutant** DNA sequence, **mRNA codon change** from **GAG** to **GUG**
2. **Glutamic acid** which is **hydrophilic / polar** in normal HbA protein is being replaced to **valine** which is a **hydrophobic/non-polar** amino acid in mutant HbS protein, resulting in a different **3D conformation shape** resulting in **sickled red blood cells**

Sickle cell disease has its highest occurrence in sub-Saharan Africa. Despite the survival disadvantage of the allele HbS, HbS allele persists at more than 12.5 % in the population.

Fig. 2.3 shows regions where sickle cell anaemia is most common, which coincide with regions where malaria is. Malaria-causing parasites are introduced into the blood by mosquitos.

Fig. 2.3



Malaria prevalence $\qquad$ ***H^S* allele frequency (percent)**

(iii) With reference to Fig. 2.3, explain why HbS allele is still preserved within the population and suggest why it occurs at an unusually high frequency in Africa.

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....[4]

ANS [L2/L3] (Novel) [4]

SC: With reference to Fig.1.6 1. explain how HbS frequency preserved
2. suggest why these alleles highest

OR:

1. Explain why
- **Diploidy → HbS: recessive allele → exist in carriers (Hb^AHb^S) and passed down thru generations**
- **hence maintain pool of HbS in the population**

2. Suggest why
- **selection pressure (malaria) is the greatest; heterozygous individuals (selective adv)**
- **need to survive malaria → HbS allele is selected for (for malaria parasite can't develop in sickle cells → X infection)**
- **need to survive from SCA → HBA allele is selected for (for efficient O_2 transport at low $[O_2]$ levels)**

[Explain why]

1. Due to diploidy, recessive alleles such as HbS are **hidden / masked over** by the **dominant allele** HbA.
2. Thus, it always existed **in carriers** (Hb^AHb^S) and passed down through generations, **maintaining the gene pool of HbS in the population**

(2 marks for points 1 & 2)

[Suggest why]

3. **Selection pressure** to survive from **malaria** is greatest (shown in Fig.1.6), thus **heterozygous** individuals are at **selectively advantage**.
4. There is a need to survive from malaria where **HbS is selected** for as **malaria parasite cannot developed in sickle cells**.
5. There is also a need to survive from sickle cell anaemia (SCA) where **HbA is selected** to allow **efficient oxygen transport** around the body, especially **at low oxygen concentrations**.

(Max 2 marks for points 3 - 5)

(b) Haematopoietic stem cell (HSC), usually derived from bone marrow, peripheral blood or umbilical cord blood have shown great promise in the treatment of sickle cell disease recently. Sources of such stem cells may be autologous (i.e. using the patient's own stem cells), allogeneic (i.e. using stem cells derived from a donor) or syngeneic (i.e. using stem cells from an identical twin).

After extraction, researchers cultivate these stem cells before injecting them back into the patient. Such haematopoietic stem cell transplantation (HSCT) therapy could potentially give new hope to patients with sickle cell anaemia, although the treatment comes with its own risk towards the patients.

(i) Explain how extracted haematopoietic stem cells are cultivated before they are injected back to the patient.

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……[2]

ANS [L1] (Novel) [2]

SC: extracted SC	**cultivate & injected back to patient**	**explain how (process)**
OR:		bone marrow (diff. source) ❷ induction for proliferation ❷ in need of growth factors / inducers ❷ differentiate into blood cells

1. Obtain / extract of **blood stem cells / haematopoietic stem cells** from the **bone marrow (or any other sources named from context given)** which is then grow them in culture dish / chambers in order to alter the surface of culture chambers to **induce** cells to undergo **proliferation.**

2. Provide the necessary **growth factors** / **change the chemical composition** of the culture medium, in addition to the nutrients in the culture medium to **induce** them to become / **differentiate** into **blood cells**

OWTTE

(ii) Explain why treatment using HSC from allogeneic sources may pose some risks to the patient.

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………[2]

ANS [L2] (Novel) [2]

SC: Explain why **HSCT treatment** **pose risk to patient**
OR: **Tissue rejection by recipient patient**
Graft-vs-Host disease ❼ WBC from donor cells 'attack' recipients' host cells (new info.)

Either

1. Recipient host cells considers **donor stem cells as foreign** and
2. **rejects** by host's immune system, which then **destroy the transplanted tissue/cells**. / **OWTTE**

Or (more accurately)

1. White blood cells from donor's immune system remain within donated cells/tissues cells **recognise the recipient as foreign** and
2. **attack** the recipient's body cells, which leads to graft-vs-host disease / OWTTE

[Total: 12]

3 Fig. 3.1 shows chickens with two different feather colours in which the gene for feather colour is carried on an autosome. The gene has two alleles, one that codes for black and the other for splashed-white. When a male chicken with black feathers is mated with a female chicken with splashed-white feathers, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.

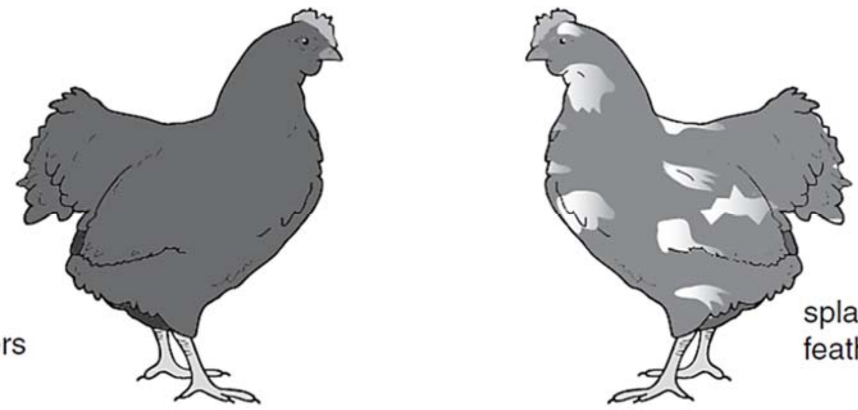

Fig. 3.1

(a) The blue feathers is the result of codominance.

Explain what is meant by 'codominance'.

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....[2]

ANS [L2] (Novel) [2]

SC: Explain what meant by codominance in this context
OR: heterozygotes❼ C^BC^W (no allele exert dominance over each other
both alleles are expressed

1. In heterozygote chickens, genotype **C^BC^W, neither allele exert dominance over the other**
2. **Both** alleles **C^B and C^W are expressed** to give the **blue colouration** on the chicken feathers.

Another gene may cause stripes on feathers (barred feathers), as shown in Fig. 3.2. This gene is carried on the X chromosome. The allele for barred feathers (**X^A**) is dominant over the allele for non-barred feathers (**X^a**).

In chickens, the male is homogametic and has two X chromosomes while the female is heterogametic and has one X chromosome and one Y chromosome.

barred feathers

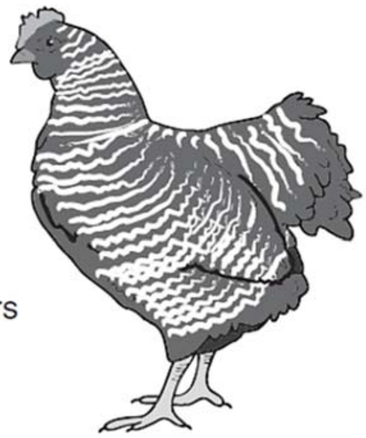

Fig. 3.2

(b) A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

Using the symbols given above draw a genetic diagram to show this cross.

parents' phenotype	male, black, non-barred feathers.	female, splashed-white, barred feathers.
genotype	$C^BC^BX^aX$	$C^WC^WX^A$ [1]
gametes	C^BX^a C^BX^a	C^WX^A C^WY [1]

C^B
$C^WX^AX^a$ [1]

offspring genotypes

phenotypes	male, blue, barred feathers.	female, blue, non-barred feathers.

ANS [L2] (H2 NYJC/2017/P2/Q7) [3]

[1] for parental genotype;
[1] for gametes;
[1] for offspring genotype and matching;

(c) Explain how a farmer could use a breeding programme to find out the genotype of a male chicken with blue, barred feathers.

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……[3]

ANS [L2] (H2 NYJC/2017/P2/Q7) [3]

SC: Explain how **genotype of male chicken** **blue barred feathers**
OR: **do test cross with female (non-barred)**

1. Carry out a **Test cross** with female with recessive trait, i.e. **non barred** feathers _ _ X^aY
2. if **all** offspring **barred** feathers, the male parent must be **X^AX^A** / **homozygous** ;
3. if **some** offspring **non-barred** features, the male parent must be **X^AX^a / heterozygous** ;

4 A student investigated respiration in a population of yeast growing in a sealed container.

Fig. 4.1 shows the results of his investigation.

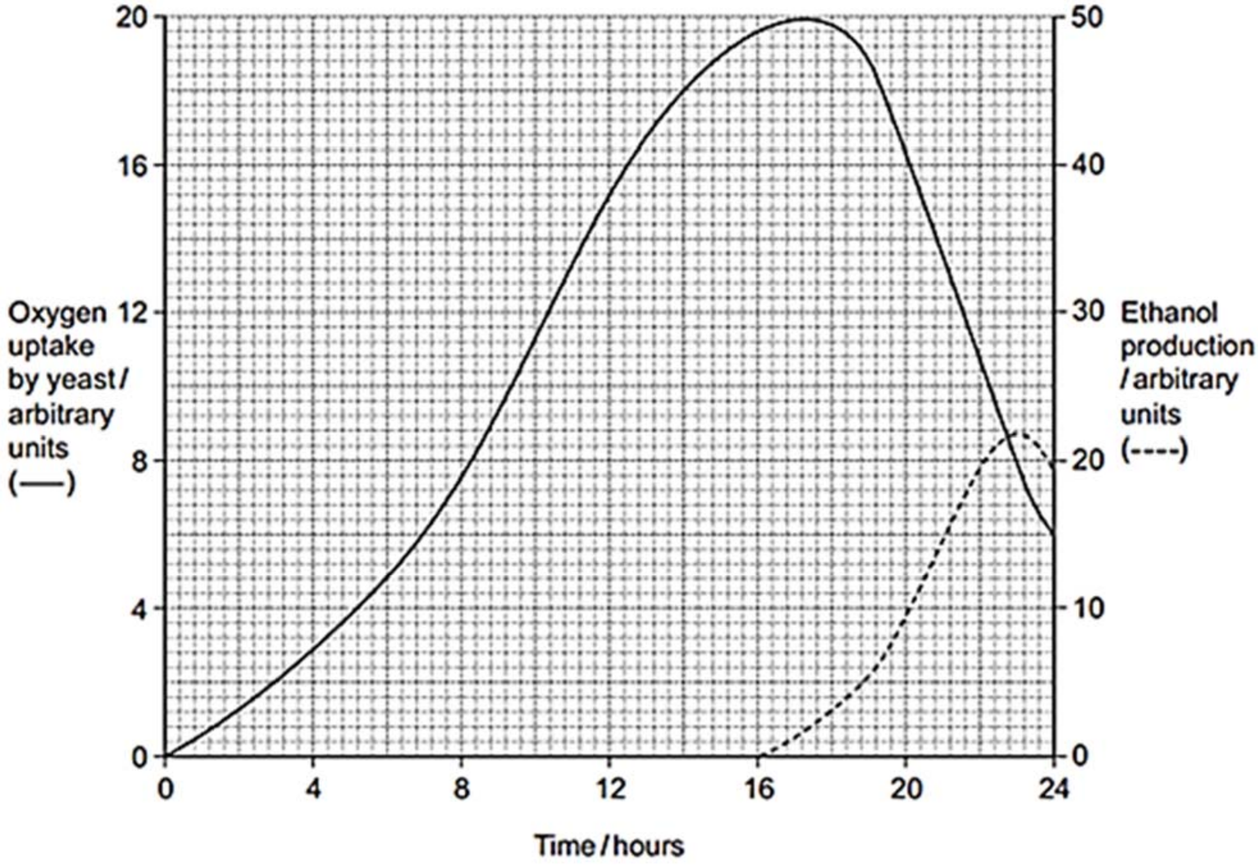

Fig. 4.1

(a) With reference to Fig. 4.1, explain the changes in the ethanol production by yeast during this investigation.

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....[3]

ANS [L2] (H2 NJC/2017/P2/Q7) [3]

SC: **With ref. Fig. 3.3** **explain changes** **ethanol production**
OR: **from 16h-24h**

[Trend]: As time ↑ from 16h to 23h, Ethanol pdtn↑ from 0-22a.u. & oxygen ↓ from 19-9a.u.

[Explain]: convert from aerobic to anaerobic respiration / ethanol or alcohol fermentation

from 23h-24h

[Trend]:As time ↑ from 23h-24h, ethanol pdtn ↓ from 22-20 a.u.

[Explain]: [glucose] ↓ / becomes limiting / ethanol reaches toxic levels

1. [*Describe*] As time **increases** from **16h to 23h**, ethanol production **increases from 0 a.u. to 22 a.u**. while oxygen concentration continue to decrease from 19 a.u. to 9 a.u.

2. [*Explain*] Yeast has convert from aerobic respiration to **anaerobic respiration** OR **ethanol / alcohol fermentation**
3. [*Describe*] As time further **increases** from **23h to 24h**, ethanol production **decreases from 22 a.u to 20 a.u.**
4. [*Explain*] This is due to **glucose concentration become limiting / decreases** OR **ethanol levels has reach toxic enough to kill the cells**

Max 3: 1 mark for describe and 2 marks for both explanation

(b) Sodium azide is a substance that inhibits the electron transport chain in respiration. The student repeated the investigation but added sodium azide after 4 hours.

Suggest how the addition of sodium azide would affect oxygen uptake and ethanol production by yeast.

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....[2]

ANS [L3] (H2 NJC/2017/P2/Q7) [2]

SC: **Suggest** **& Explain how** **sodium azide** **affect oxygen uptake**

OR: oxygen uptake ↓es / stopped ➐ **final e acceptor**

ethanol pdtn starts

switch from aerobic ➐ anaerobic / ethanol fermentation

1. Addition of sodium azide **decreases the oxygen uptake** / **causes oxygen uptake to cease / stop** since it is the **final electron acceptor** of the electron transport chain (ETC) / OWTTE

2. **Ethanol production** starts earlier as yeast switches from **aerobic respiration to anaerobic respiration / ethanol fermentation / alcohol fermentation**.

5 Monarch butterflies are one of the largest migrating butterflies in the world. They have recently come under the spotlight in research due to how global warming is affecting their populations.

Monarch butterflies, shown in Fig. 5.1, are obligate feeders on a specific species of milkweed plant for both adults and caterpillars, and lay their eggs on these plants as well. These plants also secrete a toxin called cardenolides that deter most vertebrate herbivores that the monarchs are tolerant to. The monarchs even accumulate these toxins in their bodies that make them taste bad for their predators.

Fig. 5.1

(a) Based on the information above, identify two selection pressures for the monarch butterflies.

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....

....[2]

ANS [L1] (H1 Novel) [2]

SC: identify 2 selection pressures

OR:predation on larvae; toxins in milkweed; temperature

Any two;

1. Predation on larvae; difficulty in feeding on milkweed; toxins in milkweed; accept: temperature;

(b) Explain how monarch butterflies and caterpillars have evolved to have a defense against their predators.

....

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....

....[3]

ANS [L2] (H1 Novel) [3]

SC: Explain how **monarch butterfly/caterpillars** **evolved** **defense against predators**
OR:
1. genetic variation
2. tolerant /sequester cardenolides ❷ selective adv
3. survive & reproduce similar offsprings + ↑allele freq

1. There is **genetic variation** within the population of monarch butterflies where some are less and some are more tolerant to cardenolides.
2. Those that are **more tolerant to and are able to sequester cardenolides** are at a **selective advantage** over those that are less tolerant to and/or those unable to sequester cardenolides.
3. **Over time**, these selective advantageous individuals **survive and reproduce similar offspring** and **alleles** that confer tolerance and ability to sequester cardenolides **increase in frequency**.

(c) Monarch butterflies, like most insects, are highly sensitive to weather and climate. Based on Q_{10} theory, they depend on environmental cues, temperature in particular, to trigger reproduction, migration, and hibernation.

(i) State the meaning of Q_{10} theory and describe the correlation of this theory on insect physiology.

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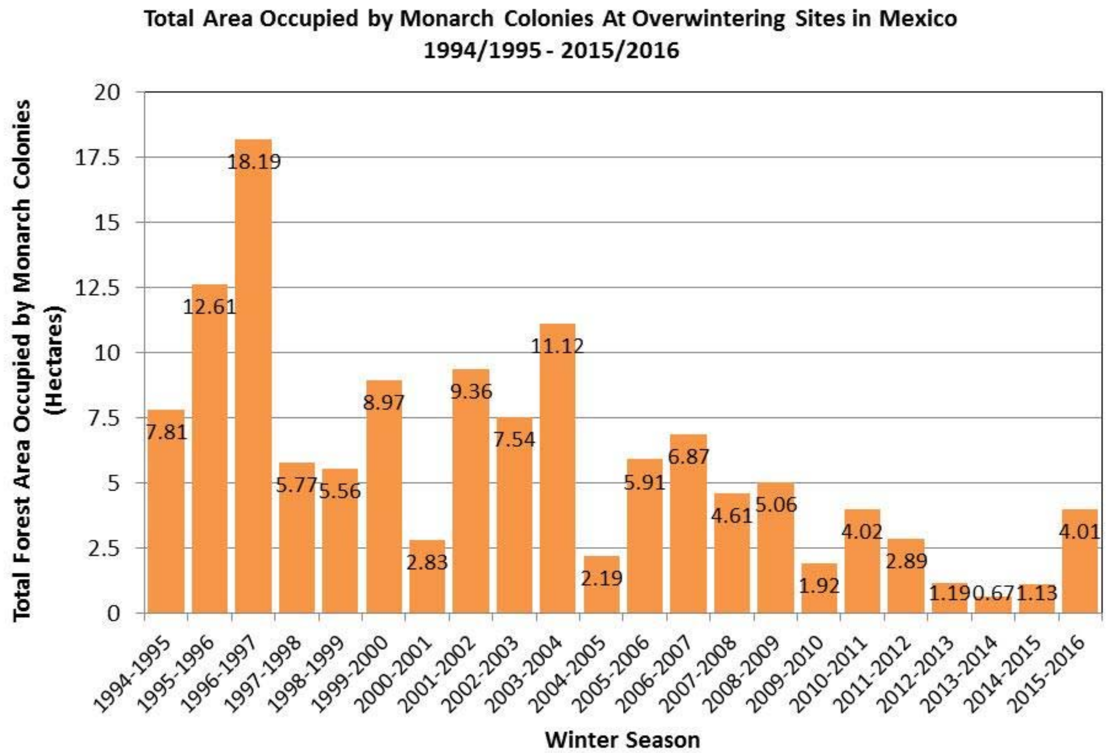
ANS [L1] (H1 Novel) [2]

SC: State	**Q_{10} theory**	**Describe correlation**
OR:	**For every 10°C ↑**	impact on insect physiology; ↑ temp → **life cycle faster; mature faster**
	2-fold ↑ in E rxn	multiply faster

1. The Q_{10} theory of enzyme states that for **every 10°C increase in temperature**, there will be **2 fold increase in the enzyme reaction / OWTTE**
2. This has an impact in insect physiology as their **lifecycle gets shorter / shorter instars**, gaining maturation stage **faster** as well as **multiply faster / breed more**. / **OWTTE**

Fig 5.2 below shows data of monarch butterflies arriving at Mexico during winter.



Data from 1994-2003 were collected by personnel of the Monarch Butterfly Biosphere Reserve (MBBR) of the National Commission of Protected Natural Areas (CONANP) in Mexico. Data from 2004-2016 were collected by the WWF-Telcel Alliance, in coordination with the Directorate of the MBBR.

Fig. 5.2

(ii) Explain why monarch butterflies would be highly sensitive to climate change.

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....[2]

ANS [L2] (H1 Novel) [2]

SC: Explain why **monarch butterflies** **highly sensitive**

OR:
1. not able to regulate temp. / depend on ambient temp
2. small size ❷ temperature diff affect them (even at small changes)
3. dependent on temp as signal to reproduce, migrate, hibernate
4. any changes in temp (esp. ↑temp due to temp) would change the cue

1. Monarch butterflies, like most insects, are cold-blooded animals; they are **unable to regulate their temperature** and **depend on ambient temperature**
2. They are **small sized**, hence **small fluctuations in temperature affect them** more than other animals in terms of metabolism
3. They depend on a **particular temperature** as a signal / cue for reproducing, migrating and hibernate, hence **global warming would change the cue** for them to carry out such crucial processes
4. **AVP**

Max 2

(iii) Based on recent research, it has been found that increasing temperature can cause milkweeds to secrete a significantly higher level of cardenolides than normal.

With reference to Fig. 5.2 and this information, suggest how climate change is likely going to affect the monarch butterflies.

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....[3]

ANS [L3] (H1 Novel) [3]

SC: with reference to Fig. 4.2 + info **suggest how** **climate change ❷ affect monarch butterfly**
OR: **[cause]** **[effect]**

1. Increase in temperature will affect the **timing of their migration**, evident by the number arriving at Mexico as **changes in temperature affects the cue to when they start migrating**;
2. Climate change will **shift the migration range of monarch butterflies** more north and higher due to increasing temperature;

3. **but the milkweed which they depend on may not be able to redistribute as quickly** since they are plants, hence resulting in loss of food;

4. Increase in temperature due to global warming may also cause the **milkweed plants to produce more toxins faster than the monarch butterfly population can adapt / evolve,** killing off some of them;

AVP (1m max)

5. Global warming **may also directly kill off** monarch butterflies due to their **narrow physiological tolerance** range to temperature

6. **AVP**

[Total: 12]

Section B
Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

6 In photosynthesis, glyceraldehyde 3-phosphate (3-carbon sugar) is produced in the stroma of chloroplast during Calvin cycle.

(a) Outline this process and how the glyceraldehyde 3-phosphate is converted into starch and discuss the suitability of starch as a storage compound in plants. [8]

(b) Compare the processes of Calvin cycle to that of the Krebs cycle. [7]

[Total: 15]

7 **(a)** Outline the process of DNA replication and discuss the suitability of DNA as a hereditary material. [8]

(b) Compare the processes of DNA transcription to that of DNA translation. [7]

[Total: 15]

6 In photosynthesis, glyceraldehyde 3-phosphate (3-carbon sugar) is produced in the stroma of chloroplast during Calvin cycle.

(a) Outline this process and how the glyceraldehyde 3-phosphate is converted into starch and discuss the suitability of starch as a storage compound in plants. [8]

ANS [L1/L3] (H1 Alevel/2009/P2/Q5c)
SC: 1. outline GALP made 2. GALP ❼ Starch 3. Discuss suitability of starch (as storage cpd)
OR:

Outline how Glyceraldehyde 3-phosphate is made in Calvin cycle:

1. In carbon fixation, **one molecule** of **carbon dioxide** is incorporated to a **5C ribulose bisphosphate** (RuBP) to give a **6C** intermediate,
2. This is **catalysed by** the enzyme **ribulose bisphosphate carboxylase oxygenase** / **RUBISCO**
3. This **6C** intermediate is **unstable**; hence it is broken down to give two molecules of **3C glycerate-3-phosphate** (**GP / PGA**).
4. In carbon reduction, **glycerate-3-phosphate** (GP) in the presence of **ATP** and **reduced NADP** / **NADPH** (both from the light dependent stage) is **reduced** to **Glyceraldehyde-3-phosphate** (**TP / GALP**).

Outline how Glyceraldehyde 3-phosphate is made in Calvin cycle and converted to starch:

5. **Glyceraldehyde-3-phosphate** (GALP) can be **combined** and **rearranged to form glucose**,
6. **Glucose monomers** undergo **condensation** to form **amylose** via **α1-4 glycosidic bond** and **amylopectin** via **α1-4 and α1-6 glycosidic bond**
7. **AVP**

Discuss starch suitability as a storage compound in plants

8. Starch has **compact shape** where **many glucose residues** can be stored in a **small volume** within the cell.
9. **Insoluble** in water therefore is a useful storage material because it has little effect on the **water potential** of cellular fluid.
10. **AVP**

All 3 components must be present for full marks

(b) Compare the processes of Calvin cycle to that of the Krebs cycle. [7]

ANS [L2] (H2 Alevel/2008/P2/Q8)
SC: compare **Calvin vs. Krebs**
OR: similarities & differences

[Max 4 for either similarities or differences]

	Features	**Calvin cycle**	**Krebs cycle**
	Similarities		
S1	Both processes require **enzymes** to drive / catalyse substrate based reactions		
S2	Both processes requires **electron carriers**		
S3	Both processes requires a **regeneration of starting material**		
S4	Both processes needed a **substrate** to start the cycle process		
S5	**AVP**		
	Differences		
D1	Location	Occurs at the **stroma** of chloroplast	Occurs at the **mitochondrial matrix**
D2	Starting material	**Ribulose bisphosphate** (RuBP) is the starting material of the cycle and eventually regenerated	**Oxaloacetate** is the starting material of the cycle and eventually regenerated
D3	Substrate	**Carbon dioxide** is used as the substrate	**Acetyl-CoA** is used as the substrate
D4	Role of CO_2	**Carbon dioxide is needed for carbon fixation.** It converts RuBP to form unstable compound 6C which eventually breaks down to glycerate-3-phosphate (GP)	**Carbon dioxide is released** as a result of decarboxylation reactions
D5	Electron carriers	**NADPH** is needed to reduce glycerate-3-phosphate (GP) to triose phosphate (TP) / glyceraldehyde-3-phosphate by serving as electron donors	**NAD+ and FAD** are required for oxidation of intermediates of the cycle by serving as electron acceptors
D6	ATP	Needed energy through **hydrolysis of ATP**	**ATP synthesis** through **substrate level phosphorylation**
D7	Products	For **every 3 molecules of CO_2, one GP** is made	Each cycle gives rise to **2 ATP, 6 NADH, 2 $FADH_2$**
D8	Process type	**Anabolic** due to synthesis of hexose sugar phosphates and eventually energy storage polysaccharide	**Catabolic** due to oxidation of Acetyl-CoA through series of decarboxylation and dehydrogenation
D9	**AVP**		

7 **(a)** Outline the process of DNA replication and discuss the suitability of DNA as a hereditary material. [8]

ANS [L1/L3] (Novel)
SC: 1. outline DNA replication process **2. Discuss suitability of DNA (as a hereditary material)**
OR:

Outline DNA replication process (Max 4):

1. **Helicase unwinds DNA double helix** and **unzips** by **breaking hydrogen bonds between complementary base pairs** to produce two template strands where **single stranded binding proteins** keep the **template strands apart;**
2. **Primase binds** to template at **3'ends** of template strand and **synthesize primers** and provide **free 3'OH for initiation of replication** by DNA polymerase;
3. **DNA polymerase III** carries out DNA **elongation** in **5' to 3'** direction, **catalysing** formation of **phosphodiester bonds** between incoming DNA nucleotides;
4. **DNA nucleoside triphosphates added** to the template strand by **complementary base pairing**
5. **Continuous replication** on **leading strand**;
6. **Discontinuous replication** on **lagging strand** in the form of **Okazaki fragments;**
7. **DNA polymerase I removes RNA primers** and **replaces them with DNA nucleoside triphosphates;**
8. **DNA ligase** formed **phosphodiester bonds between DNA nucleotides** of Okazaki fragments**.**

DNA molecule for role as a stable molecule for inheritance (Max 4)

9. **Strong covalent phosphodiester bonds** between adjacent **nucleotides** of the same DNA strand / within each strand;
10. **Hydrophobic interactions between** the **stacked bases**;
11. **Hydrogen bonds** between **complementary base pairs** in the **double helix / complementary nitrogenous bases** of adjacent **DNA strands;**
12. **Purine always pairs with pyrimidine**
13. **Ensures width of DNA /** between the **2 sugar phosphate backbones is constant**
14. **Stabilises** the **structure of the double helix** and m**aintains** the **integrity** of the **DNA sequence.**

DNA molecule in relation to accurate DNA replication

15. The **sequence of bases on the two strands are complementary** / nitrogenous bases allow for highly specific **base-pairing** to occur between **complementary bases**;
16. **Adenine** pairs with **Thymine & Guanine** pairs with **Cytosine**;
17. **Each parental strand can act as a template to which a complementary set of deoxyribonucleotides** will attach by base pairing / synthesize a new daughter strand;
18. Therefore each original / parental **DNA molecule** can give rise to **2 copies** which are **identical** in structure and base sequence;
19. Allow for **accurate replication** to occur; for transmission to daughter cells / next generation

In terms of DNA repair

20. The sequence of bases on the two strands are complementary / As nitrogenous bases allow for highly specific base-pairing to occur between complementary bases;

21. Adenine pairs with **Thymine; Guanine** pairs with **Cytosine**

22. Intact / existing **complementary strand** can be used as a **template** to guide **DNA repair**;

23. Repair mechanisms ensure that the **integrity of the DNA sequence remains intact;**

[Note: to credit only once for the same marking point.]

(b) Compare the processes of DNA transcription to that of DNA translation. [7]

ANS [L2] (Novel)
SC: compare **DNA transcription Vs. DNA translation**
OR: similarities & differences

[Max 4 for either similarities or differences]

	Features	**Transcription**	**Translation**
	Similarities		
S1	Both processes require a **template**		
S2	Both processes are **catalyzed by enzymes**		
S3	Both processes involve **complementary base-pairing**		
S4	Both processes **requires energy for bond formation during elongation**.		
S5	Both processes **form polymers as product**.		
S6	**AVP**		
	Differences		
D1	Location	In **nucleus**	At **Ribosomes**
D2	Template	**DNA** template/coding strand	**mRNA**
D3	Key enzyme	**RNA polymerase** catalyses formation of phosphodiester bond between adjacent ribonucleotides	**Peptidyl transferase** catalyzes the formation of peptide bond
D4	Bond between monomers	**Phosphodiester bond** is formed **between the 5'-phosphate group** of one nucleotide **and 3'-OH** of ribose of the next nucleotide	**Peptide bond is formed between carboxyl group** of one amino acid **and the amino group** of the next amino acid
D5	Monomers (substrate)	**Ribonucleoside triphosphate**	**Amino acids** attached to tRNA
D6	Product(s)	**mRNA, rRNA and tRNA**	**Polypeptide**
D7	Fate of product(s)	products **exit nucleus and migrate to the cytoplasm**	Polypeptide chain **remain in the cytoplasm or secreted out of the cell**
D8	**AVP**		

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